A point estimate needs a loss function

After observing y, estimate the latent state $z \in \mathbb{R}^L$ by a vector $a \in \mathbb{R}^L$.

Assume $E[\|z\|^2 \mid y] < \infty$.

Squared-Euclidean loss and conditional risk

$$L(z, a) = \|z - a\|^2$$

$$R(a|y) = E[L(z, a)|y] = E[\|z - a\|^2 \mid y]$$

- Conditional risk averages the loss over the posterior $p(z|y)$.

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Center the error about the posterior mean

$$m = E[z|y] \quad \big(= \mu_{z|y} \ \text{ for the Gaussian}\big)$$

$$z - a = (z - m) + (m - a)$$

$$\|z - a\|^2 = \|z - m\|^2 + 2(z - m)^T(m - a) + \|m - a\|^2$$

Centered residual $z - m$ + displacement $m - a$.

- Conditional risk averages the loss over the posterior $p(z|y)$.